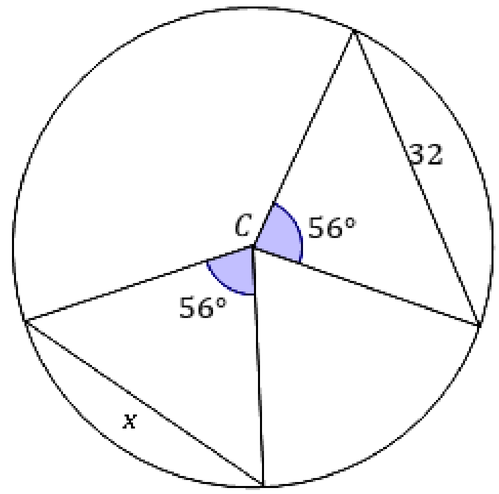
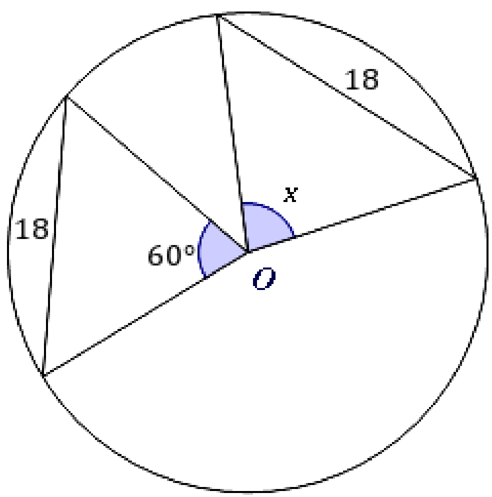
Extension: Chords of a circle

1 Find x for the following diagram with C as the center of a circle:

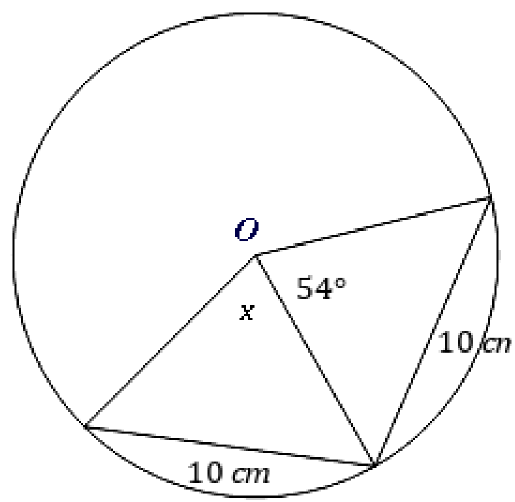


2 Find x for each of the following diagrams with O as the center of the circles:

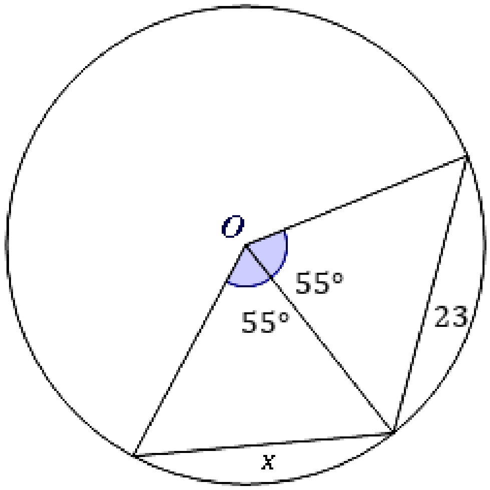
a

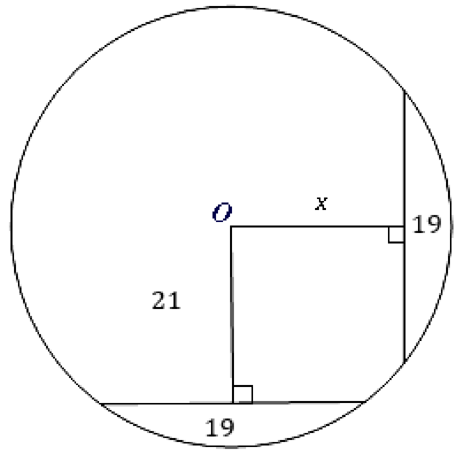


b

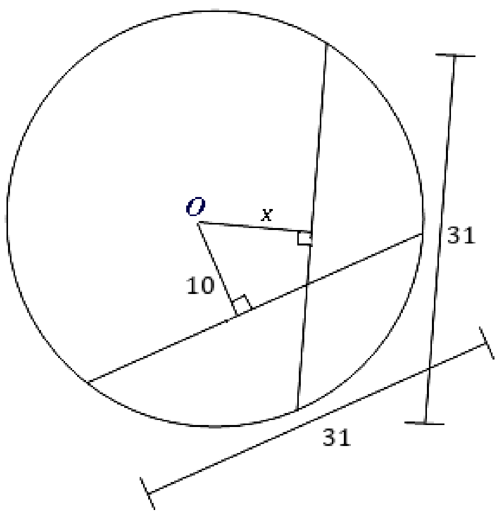


c

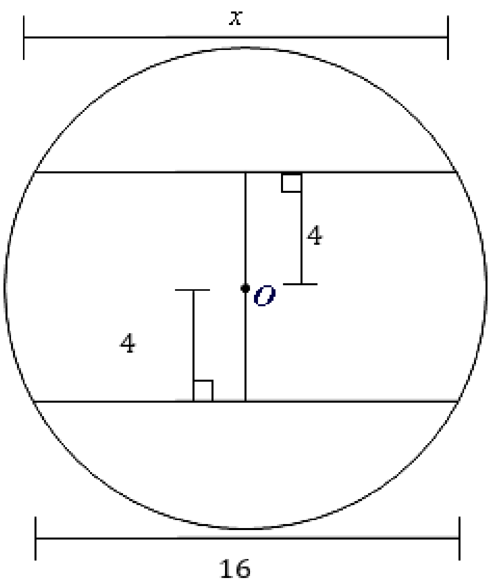




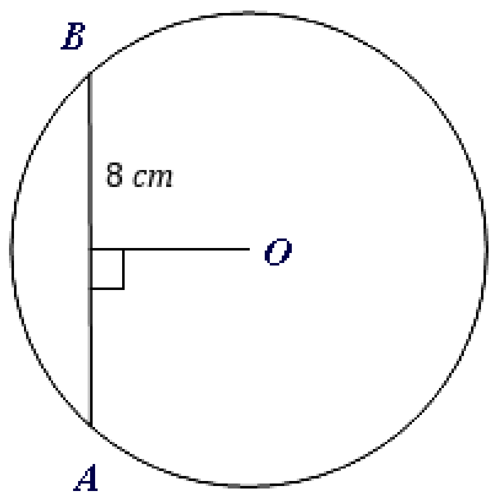
e



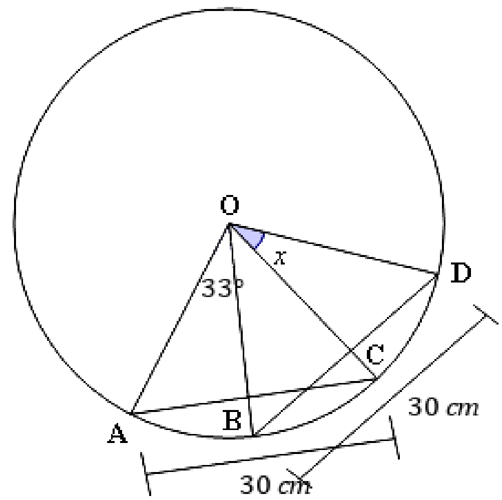
f



3 Consider circle O shown.
Find AB.

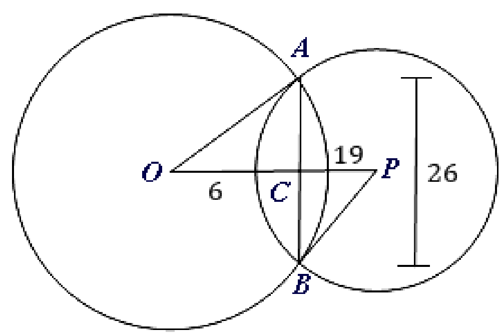


4 In the diagram, O is the center of a circle with $m\angle AOB = 33^\circ$ and $m\angle COD = x$. Find x.

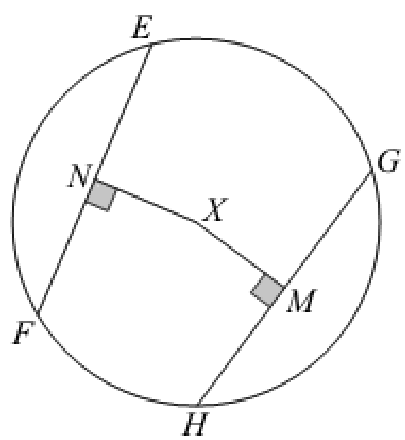


5 In the diagram, O is the center of a circle with $OC = 6$, $CP = 19$ and $AB = 26$.

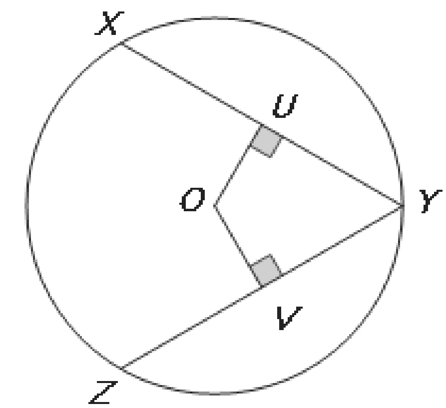
- **a** Find the radius OA. Give your answer as an exact value in radical form.
- **b** Find the radius PB. Give your answer as an exact value in radical form.



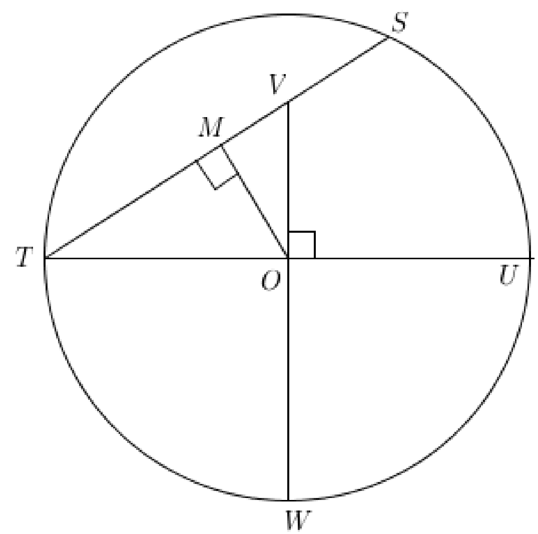
6 In the diagram, X is the center of a circle with $EF = GH$ and $XM = 9$ cm. Find XN.



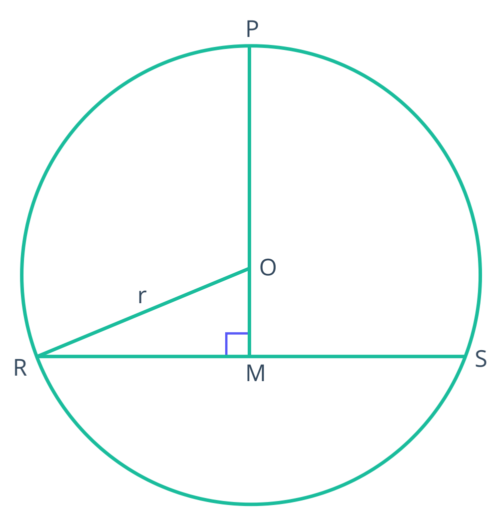
7 In the diagram, O is the center of a circle with $OU = OV$, and $XY = 19 \text{ cm}$.
Find YZ.



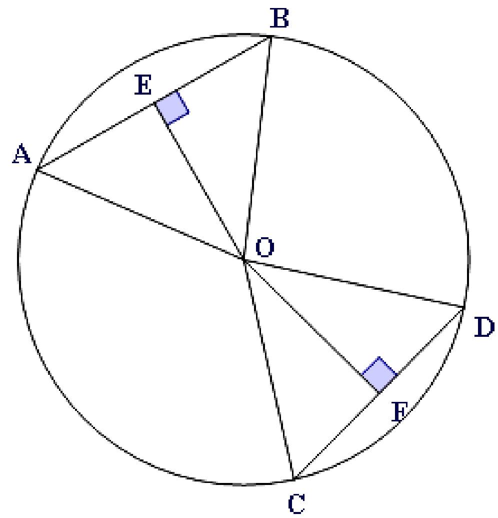
8 In the diagram, O is the center of a circle with $TU = 150 \text{ cm}$, $VW = 131.25 \text{ cm}$ and $OM = 45 \text{ cm}$.
Find SV.



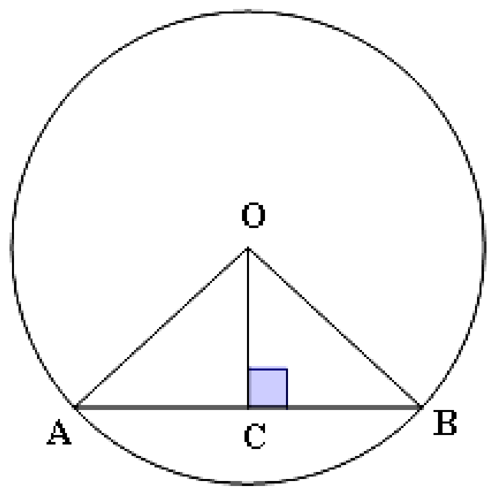
9 The diagram shows a circle with center O and $\overline{RS} \perp \overline{OM}$. The length of the chord $\overline{RS}$ is 16 cm and $\overline{OM}$ is 15 cm.
Find the length of the radius.



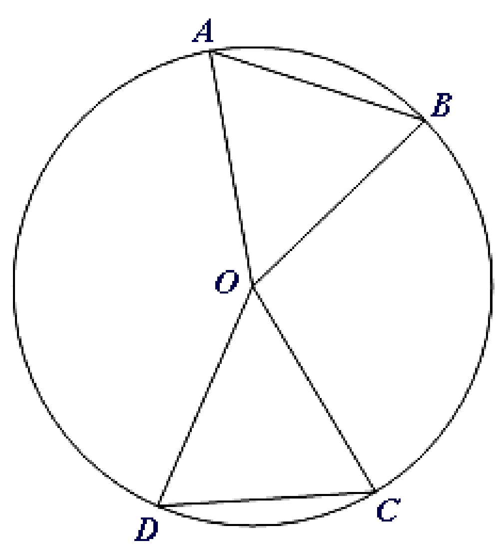
10 In the following diagram, O is the center of a circle with $AB = CD$ and $m\angle OEB = m\angle OFD = 90^\circ$.
Prove that $\triangle ABO \cong \triangle CDO$.



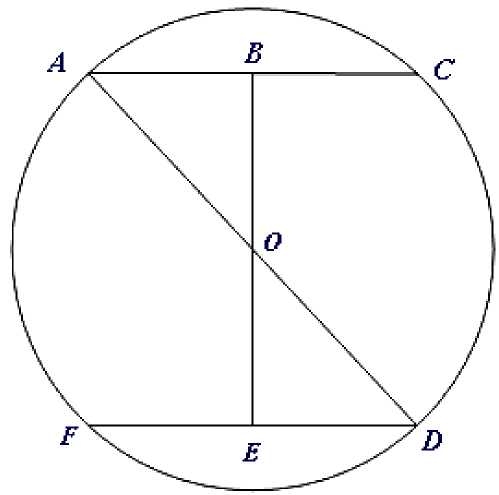
11 In the given diagram, O is the center of a circle with $\overline{OC} \perp \overline{AB}$.
Prove that $\triangle OAC \cong \triangle OBC$.



12 The diagram shows O as the center of the circle and $\angle AOB \cong \angle DOC$.
Prove that $\overline{AB} \cong \overline{DC}$.



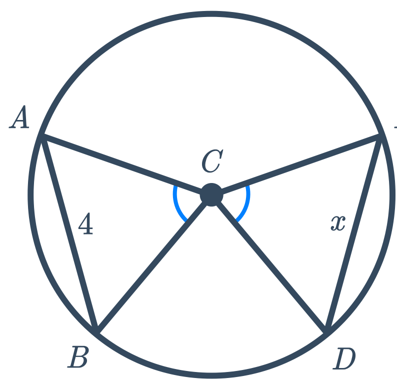
13 Consider the following circle of center O with $\overline{AB} \parallel \overline{ED}$.
Prove that $\triangle ABO \cong \triangle DEO$.



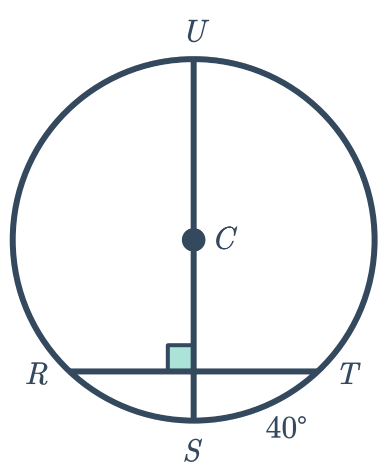
Additional questions

14

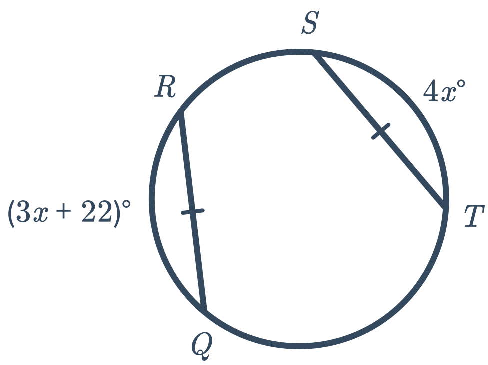
a Solve for x

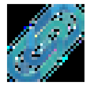


b Solve for $m\overset{\frown}{RS}$

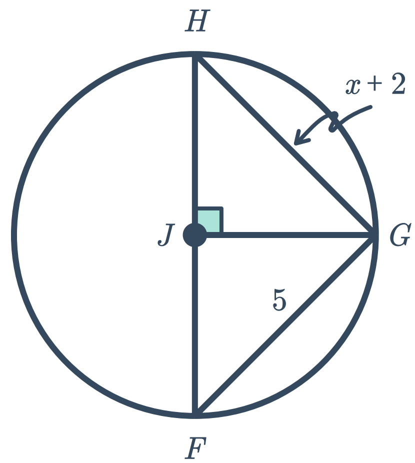


c Solve for $m\overset{\frown}{RQ}$

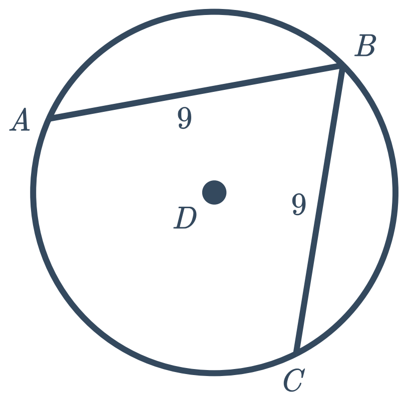




Solve for x

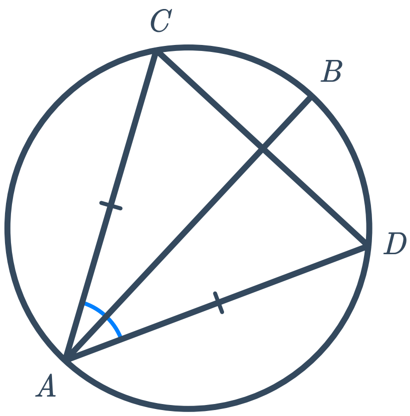


15 If $m\overset{\frown}{AB} = 118^\circ$, solve for $m\overset{\frown}{BC}$.

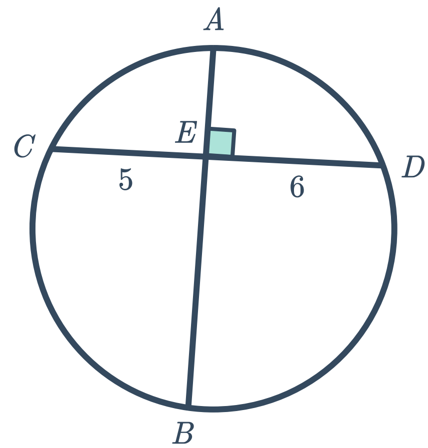


16 Based on the information given in the diagr[redacted]etermine whether each of the following statements is true or false:

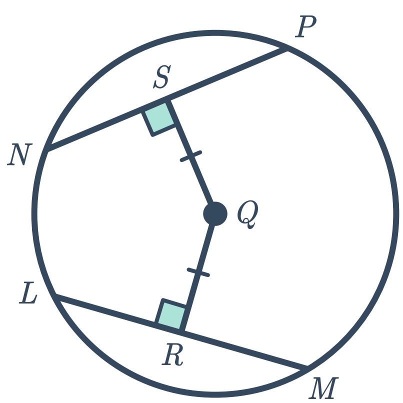
a $\overset{\frown}{AC} \cong \overset{\frown}{AD}$



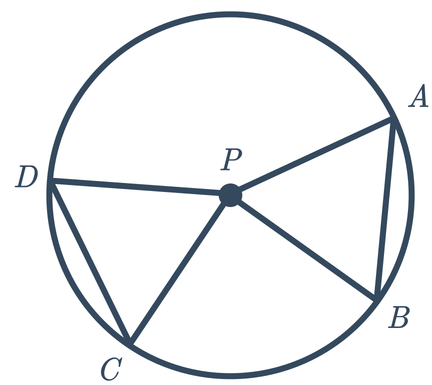
b $\overset{\frown}{AC} \cong \overset{\frown}{AD}$



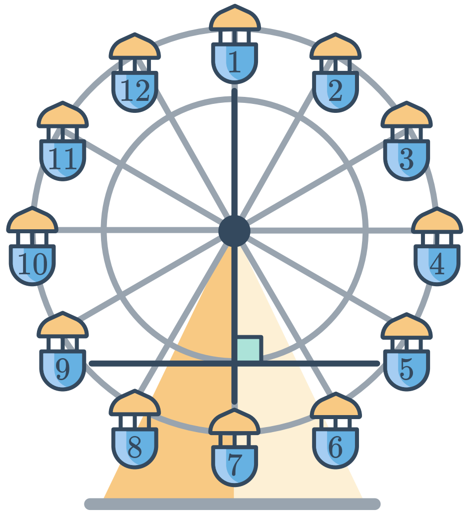
c $\overline{QS} \cong \overline{SP}$



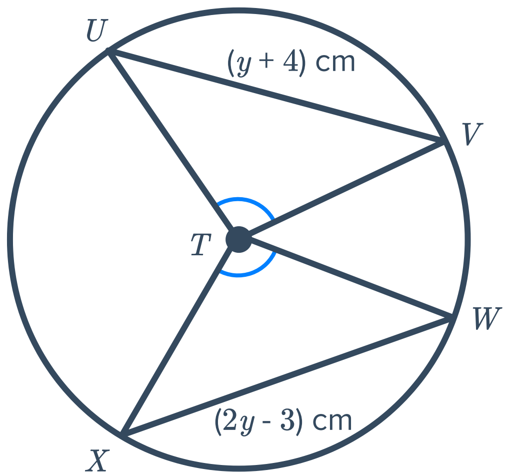
d $\angle DPC \cong \angle APB$



17 The beam connecting Cart 1 and Cart 7 on t[redacted]rris Wheel passes through the center of the wheel, forming the diameter of the circle. If the distance from Cart 5 to Cart 9 is 60 feet, how far is it from Cart 9 to the diameter of the ferris wheel?



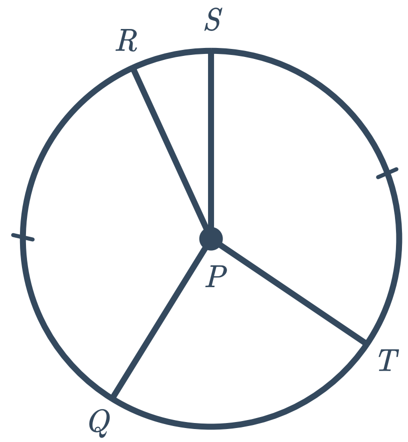
18



a Solve for y

b Solve for UV

19 $m\angle RPQ = (8x + 16)^\circ$ and $m\angle SPT = (12x$ [redacted]$^\circ$:

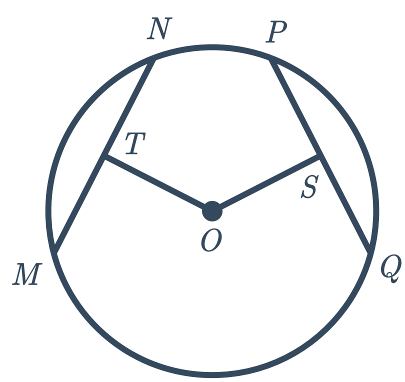


a Solve for x

b Solve for $m\angle SPT$

20 Given:

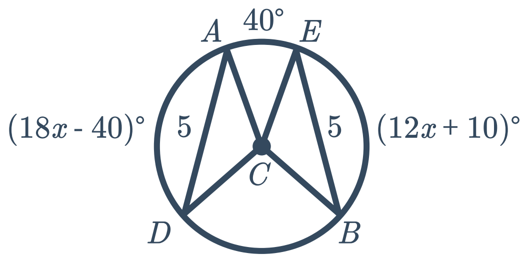
- circle O
- $\overline{MN} \cong \overline{PQ}$
- $MN = 7x + 13$
- $PQ = 10x - 8$



a Solve for x

b Solve for MN

21 Given circle C.

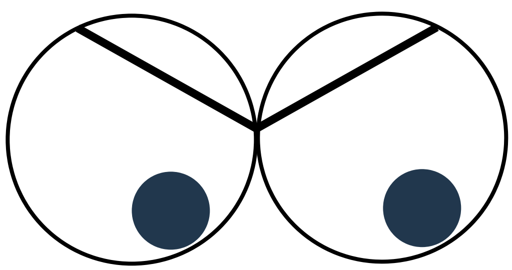


a Explain why $\angle ACD \cong \angle ECB$

b Solve for $m\overparen{AD}$

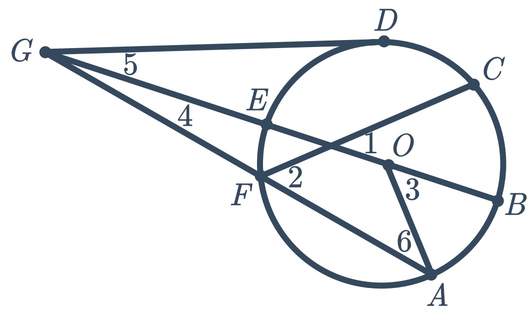
c Solve for $m\overparen{DB}$

22 Ida enjoys drawing cartoon eyes consisting of two circles the same size. She has noted that when drawing angry eyes, it looks best when the eyebrows cut through the eyes so that the measure of the arc above the eyebrows are the same length. What does this mean about the length of the eyebrows?



23 Given:

- $m\angle AGB = 12^\circ$
- $m\overset{\frown}{AB} = (2x)^\circ$
- $m\overset{\frown}{EF} = 44^\circ$
- $m\overset{\frown}{CB} = (3x - 47)^\circ$
- $\overline{EB}$ is a diameter of circle O.



a Solve for x

b Solve for $m\angle 1$

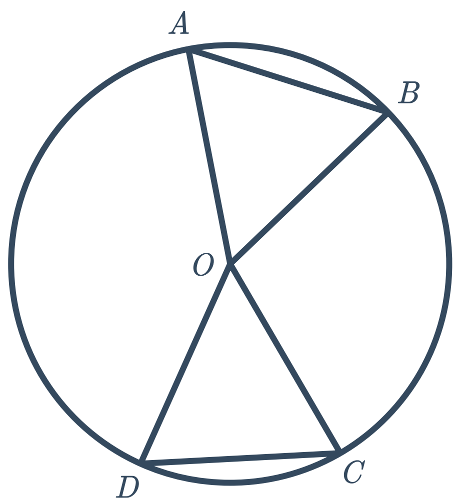
c Solve for $m\overset{\frown}{EDC}$

d Solve for $m\angle 2$

24 Given:

- circle O
- $\angle AOB \cong \angle DOC$

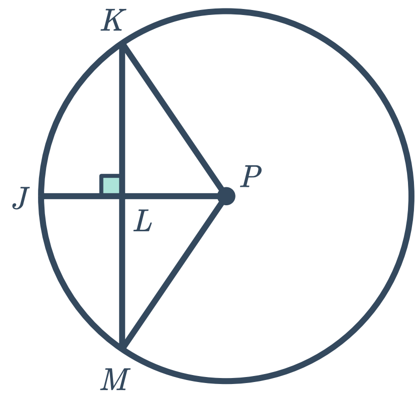
Prove: $\overset{\frown}{AB} \cong \overset{\frown}{DC}$



25 Given:

- circle P
- $\overline{KM} \perp \overline{JP}$

Prove: $\overline{JP}$ bisects $\overset{\frown}{KJM}$



26 If $\overset{\frown}{BC} \cong \overset{\frown}{ED}$, explain why $\overline{CD} \cong \overline{EB}$